Health economics: What are the theoretical predictions regarding hospitals' behavioral responses to quality-linked payments?

Octave de SAINT-MARTIN
Ahmed-Zakaria HEDROUG
Cindy JMILI
Lucas SIMATOVIC

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1 Introduction: FFS vs PPS vs VBP

Fee-for-Service (FFS) in the Retrospective Payment System

Under retrospective Fee-for-Service systems, hospitals are reimbursed based on actual costs or services provided. While this avoids patient selection and encourages high quality, it lacks cost-control incentives, often leading to "cost escalation" through increased service volume [1].

Prospective Payment Systems (PPS) and DRGs

Prospective Payment Systems (like DRGs) provide fixed payments per episode of care [1]. This maximizes technical efficiency and cost control but introduces risks of "under-provision" (skimping) and "selection," where hospitals avoid high-cost, severe patients to protect margins [1].

Value-Based Purchasing : the Quality-Linked Payments

VBP bridges this gap by adjusting DRG payments based on performance metrics [2, 4]. By making price a function of quality, VBP aims to align profit motives with patient welfare. However, its efficiency depends on precise risk adjustment; otherwise, hospitals may still engage in risk selection to protect their scores [2].

Payment Mechanism	Cost Control Incentive	Quality Incentive	Risk Selection Incentive
Fee-for-Service (Retrospective)	None (Cost Escalation)	High (Volume/Intensity)	Low
Prospective Payment (DRG)	Maximum	Low (Risk of skimping)	Maximum (Avoid severe cases)
Value-Based Purchasing	High	High (Targeted Metrics)	High (If risk adjustment fails)

Table 1: Comparison of Incentives across Hospital Payment Systems [1, 2, 4, 5]

2 Measured Quality, Performance, and Behavioral Responses of Hospitals

As discussed in the first part, prospective payment systems (such as DRGs) are effective at controlling costs, but they carry a major risk: they incentivize hospitals to cut corners on quality to preserve their margins (often referred to as *skimping*). It is precisely to counter this drift that "Pay-for-Performance" (P4P) models, such as the *Hospital Value-Based Purchasing* (VBP) program, were created [2].

These programs aim to resolve the agency problem by linking remuneration to observable quality indicators. However, the risk is to create perverse incentives: to obtain their bonuses, providers may neglect unmeasured tasks, select only low-risk patients, or embellish the numbers [2].

2.1 The Metrology of Quality: From Process to the Ceiling Effect

The *VBP* program relies on a redistribution mechanism where hospitals fund their own quality bonuses by foregoing a fixed portion of their revenue (ex 2%). This sum is then redistributed back based on their merit, calculated via a comprehensive score: the *Total Performance Score* (TPS) [4].

Based on the Donabedian approach, this score reflects the real quality of service by crossing four key aspects: safety, efficiency, clinical outcomes, and the patient experience (HCAHPS). To capture quality, most modern programs return to the fundamentals laid out by Avedis Donabedian (1966), who decomposes care into three interdependent levels [3, 4, 6].

- **Structure (The Potential):** This is the inventory of human and technological capital. It is the foundation: understanding how the hospital produces care while taking quality into account. However, there is a limit: a hospital may possess cutting-edge equipment without necessarily knowing how to use it efficiently.
- **Process (The Action):** It is not about knowing if the patient is cured, but verifying if the medical team scrupulously applied recognized scientific protocols. This is the payment systems' preferred lever because it reassures clinicians (does not depend on external factors or randomness). Nevertheless, once standardized and when 95–99% of hospitals achieve compliance, these measures suffer from the "ceiling effect": the indicator becomes useless for differentiating performance, forcing the regulator to constantly seek new indicators [2].

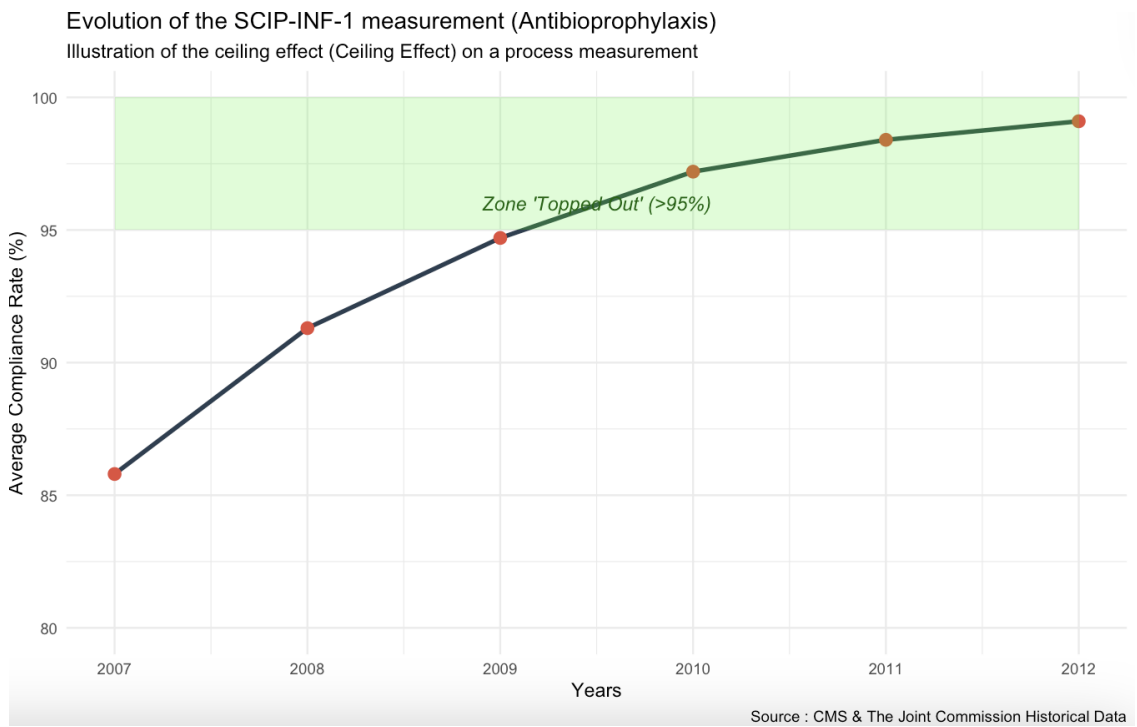

Figure 1: **The Ceiling Effect on the SCIP-INF-1 measure.**
Based on historical CMS data (2007-2012) [11]. We observe that average compliance rose from 85.8% to 99.1% in five years. Once in the "green zone" (¿95%), the measure loses its ability to differentiate between hospitals (zero variance), validating the rapid obsolescence of process measures predicted by theory [2].

- **Outcomes (The Real Value):** This is the value created, what truly matters to the patient: did they survive? Did they return to the hospital (readmission)? How did they experience their stay? Unfortunately, outcomes can be hindered and biased by other variables, often exogenous factors or externalities not controlled by the hospital (patient age, social precariousness).

2.2 Strategic Behavioral Responses: Multitasking and Selection

The "Multitask" trap (Teaching to the test)

To understand this phenomenon, we must return to the **Holmström & Milgrom (1991)** model [9]. The relationship between the payer (State/Insurer) and the hospital is marked by strong information asymmetry. Since the agent's effort

is not observable by the payer, the payer is considered "blind." To compensate, they write a contract based on what they can see: numbers and indicators. The hospital, seeking to maximize its utility while minimizing its costs, will reallocate its efforts only towards remunerated tasks. Essential but invisible or unmeasured tasks consequently risk being neglected. This phenomenon is named *"Teaching to the test"*.

The HRRP Program Example (Gupta et al., 2018) [7] This mechanism of effort reallocation can have unexpected side effects. To avoid penalties linked to readmissions (a measured task), some hospitals delayed the return of heart failure patients to the emergency room. The financial objective was achieved, but this may have occurred to the detriment of an indicator unmonitored at that time: mortality.

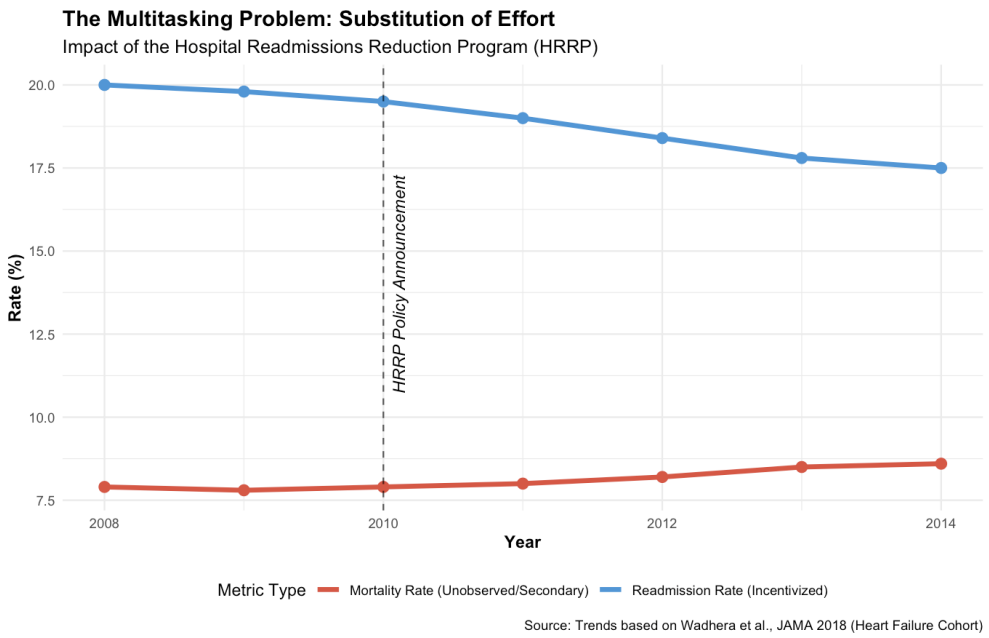

Figure 2: **Empirical evidence of "Teaching to the test" (HRRP Program).**
This graph illustrates the perverse effects predicted by Holmström & Milgrom [9]. Following the announcement of penalties in 2010 (dotted line), hospitals concentrated their efforts on reducing readmissions (blue curve), as this was the financially sanctioned metric. However, this effort came at the expense of overall clinical surveillance, leading to a concomitant rise in mortality (red curve) [8].

If the hospital cannot improve its results by working more, it has another rational option: changing its patients. This is :

Adverse Selection and Cream Skimming

They will focus on low-risk patients, thereby modifying results in their favor by omitting riskier patients susceptible to lowering their statistics [2].

The study by **Hofer et al. (1999)** [10] on diabetes is a perfect illustration. They showed that the level of glycated hemoglobin (HbA1c) often depends more on the social situation of the patient than on the doctor's talent. More disturbingly, they proved via simulation that the most efficient method for a doctor to improve their ranking was not to provide better care to their entire patient base, but to "untick" or exclude from their panel 1 to 3 patients who presented the highest rates. By removing these few complex patients, often the most socially vulnerable, a provider could move from mediocre to elite performance.

Administrative Upcoding

Finally, hospitals can resort to Upcoding: maximizing the documentation of minor comorbidities to increase the patient's "expected severity." If the patient survives, the hospital appears more performing. This is only an administrative improvement [1].

3 Critical evaluation of the economic efficiency of quality-linked payment mechanisms

The transition to quality-related hospital payment systems such as the Hospital Value-Based Purchasing (HVBP) program represents an attempt to correct a major limitation of traditional payment arrangements. These systems have often failed to properly align hospitals' financial incentives with social objectives. Evaluating the economic efficiency of quality-linked payments therefore requires considering both their theoretical functioning and their practical limitations, and comparing them explicitly with fee-for-service and prospective payment systems.

3.1 Potential efficiency of quality-linked payments

Theoretically, quality-linked payment policies are expected to improve economic efficiency by addressing incentive problems present in standard hospital payment systems. By linking part of hospital reimbursement to quality and performance indicators, these mechanisms aim to better align financial incentives with the healthcare system's objectives [2].

First, quality-linked payments can improve the alignment between incentives and objectives. Unlike fee-for-service payments, which mainly reward the volume of care provided, or prospective payment systems, which primarily encourage cost reduction, quality-linked payments attempt to internalize both cost and quality considerations [1]. Hospitals are therefore encouraged to provide a more efficient level of care, either by improving quality for a given cost or by maintaining quality while reducing costs.

Second, quality-linked payments may help reduce preventable events by emphasizing indicators related to clinical outcomes, such as mortality, complications, patient safety, and patient experience. Reducing medical errors, avoidable complications, or hospital readmissions can improve patient welfare while also lowering unnecessary healthcare expenditures, leading to potential gains in allocative efficiency [2].

Finally, quality-linked payments are often designed to complement prospective payment systems. In the case of the HVBP program, quality incentives are introduced through bonuses and penalties funded by a withholding of existing payments rather than replacing prospective payment per hospital stay [4]. This structure allows hospitals to maintain incentives for cost control while limiting the risk of quality deterioration associated with prospective payments alone [1].

3.2 Limitations and risks of inefficiency

Despite these potential benefits, the economic efficiency of quality-linked payments depends heavily on their design and implementation. Several limitations may reduce their effectiveness or even generate new sources of inefficiency [2].

A major limitation is the imperfect measurement of quality. Quality of care is multidimensional and partly unobservable, and the indicators used capture only certain aspects of hospital performance. As a result, hospitals may focus their efforts on measured dimensions of quality while neglecting important but unmeasured aspects. This reallocation of effort, often referred to as a multitasking problem, can reduce overall efficiency [2, 9].

In addition, implementing quality-linked payment programs involves significant administrative costs. Data collection, reporting, and verification require substantial financial and human resources. If the gains from quality improvements are limited, these administrative costs may offset part of the expected efficiency gains [2].

Quality-linked payments may also increase disparities between hospitals. Facilities treating more complex or disadvantaged patient populations may obtain lower performance scores, especially when risk adjustment is imperfect. These hospitals may therefore face financial penalties, which can limit their ability to invest in quality improvement and reduce overall system efficiency [2].

Finally, the effectiveness of these mechanisms depends on the size of the financial incentives. When the share of payment linked to quality is small, hospitals may respond only weakly to these incentives and improvements in quality and efficiency remain limited, while the costs of implementing the program persist [2].

3.3 Synthetic comparison with fee-for-service and prospective payment systems

To better assess the economic efficiency of quality-based payments, it is useful to compare them directly with traditional hospital payment systems. Fee-for-service payments mainly encourage an increase in the volume of care, while prospective payment systems provide stronger incentives to reduce costs per hospital stay, sometimes at the expense of quality. Quality-based payment mechanisms aim to add an additional incentive focused on performance and outcomes [1].

4 References

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AI use in the Health Economics Assignment made by Octave de SAINT-MARTIN, Ahmed-Zakaria HEDROUG, Cindy JMILI, Lucas SIMATOVIC

- to make graphs (graphe 1 : "Write R code to generate an academic-quality line graph illustrating the ceiling effect in hospital process quality measures. Use simulated data consistent with CMS SCIP-INF-1 compliance rates between 2007 and 2012 (approximately increasing from 86% to 99%). The x-axis should represent years, the y-axis compliance rates. Highlight the threshold at 95% to show the ceiling effect. Use ggplot2 ». For Graphe 2 : « Generate R code to plot two time-series illustrating the 'teaching to the test' effect in the Hospital Readmissions Reduction Program. The graph should show decreasing readmission rates and increasing mortality rates after the 2010 policy announcement. Include a vertical line at 2010. Use ggplot2 »)

- to make the tab (CAN you provide me a table in a Latex format 4*4 about the 4 payments mechanisms, cost control inventives, quality inventives and risk selection of the Previous définitions and their measures for each.)

- to support some of our arguments (the following: The argument that quality-linked payment systems improve economic efficiency by better aligning hospital incentives with social objectives, through the joint consideration of cost and quality (Section 3.1). And the normative comparison between fee-for-service, prospective payment systems, and value-based purchasing, focusing on their expected effects on costs, quality, and overall economic efficiency (Section 3.3).)